Project Report

Streamlining IT Procurement:
Automating Standard Laptop Orders with Flow Designer

Team ID: _____

Team Members

Team Member 1: Project lead

Team Member 2: ServiceNow developer

Team Member 3: Business analyst

Team Member 4: IT procurement coordinator

Team Member 5: Tester

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Source Code

Dataset Link

GitHub & Project Demo Link

1. INTRODUCTION

1.1 Project Overview

Ordering a standard laptop in many organizations depends on emails, manual approvals and hand-entered purchase orders. This project automates that process in ServiceNow using Flow Designer. An employee submits one catalog request. The flow checks eligibility, approves automatically for pre-approved roles or sends a one-click approval to the manager, checks stock, creates a purchase order when needed, creates the asset record and notifies the employee at every stage.

1.2 Purpose

The purpose of the project is to get approved devices to employees faster, remove repetitive procurement work from IT staff, and give everyone a clear view of request status. The goals are shorter turnaround, fewer manual steps and fewer errors.

2. IDEATION PHASE

2.1 Problem Statement

How might we automate standard laptop orders in ServiceNow using Flow Designer, so employees receive approved devices faster and IT spends less time on repetitive procurement work?

PS	I am	I am trying to	But	Because	Which makes me feel
PS-1	A new employee who needs a standard laptop	Get a laptop before my start date	The request needs many manual approvals and follow-ups	Requests are handled by email and each step waits on a person	Anxious and unproductive
PS-2	A department manager	Approve my team's requests quickly	I get requests with missing details and repeated reminders	There is no rule to pre-approve eligible roles	Overloaded and unsure about budget
PS-3	An IT procurement coordinator	Fulfil orders accurately	I re-enter details, check stock by hand and chase status	The steps are not connected in one automated flow	Frustrated and stretched

2.2 Empathy Map Canvas

Empathy maps were built for the employee (Riya) and the IT procurement coordinator (Arjun).

	Riya, requesting employee	Arjun, IT procurement coordinator
Says	"When will my laptop arrive?" "Who do I ask for an update?"	"I keep re-entering the same details." "Do we have this model in stock?"
Thinks	"Why does a standard laptop need so many approvals?"	"Most of these requests are identical."
Does	Emails IT and her manager for status; waits and follows up	Checks inventory by hand; chases approvals; creates orders manually
Feels	Anxious, frustrated, unsure the request is moving	Stretched by repetitive tasks and constant status questions
Pains	No status, long waits, repeated information	Manual hand-offs, status inquiries, errors from re-keying
Gains	Simple request, automatic approval, laptop on day one	Routine steps automated, time for exceptions, accurate records

2.3 Brainstorming

The team of five brainstormed ideas, grouped them into five themes and placed them on an importance and feasibility grid.

Theme	Ideas
Make it easy to request the right laptop	Standard catalog item with pre-filled specs; auto-fill from user record; eligibility check and duplicate blocking
Approve quickly without manual chasing	Skip approval for pre-approved roles; one-click manager approval; auto-escalation; budget check by cost center
Fulfil from stock or vendor automatically	Inventory check and reservation; auto-created purchase order; vendor assignment by region or price
Keep everyone informed	Stage-by-stage notifications; portal status tracker
Close the loop and measure results	Auto-created asset record on delivery; turnaround and savings dashboard

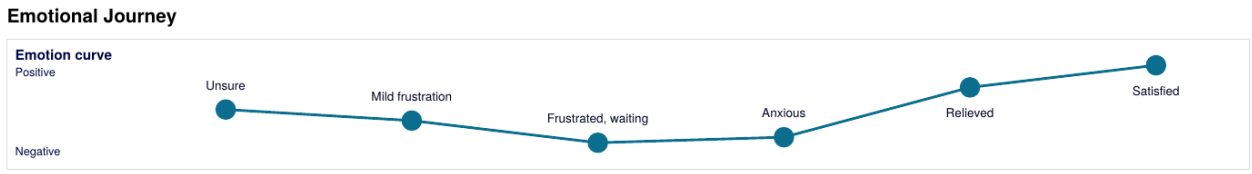
Prioritization: The ideas with high importance and high feasibility (skip approval for pre-approved roles, standard catalog item, inventory check and reservation, stage notifications) are built first. Budget check, automatic purchase order and escalation come next. Asset record, vendor assignment and the dashboard follow in a later phase.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey map follows Riya through six stages, from the need for a laptop to closing the request. The emotional low point is the approval and fulfilment stage, where she waits without visibility.

	1. Need arises	2. Submit request	3. Approval	4. Fulfilment	5. Delivery and setup	6. Close and review
Customer actions	Employee joins or needs a device replacement and finds the laptop request option	Opens the service catalog, picks the standard laptop and submits the request	Manager receives the approval task and approves or rejects it	Stock is checked; a purchase order is raised if the laptop is not in stock	Receives the laptop, signs for it and logs in	Confirms receipt and gives feedback on the process
Touchpoints	Employee portal, onboarding checklist	Service catalog form	Email and mobile approval	Inventory records, vendor order	Delivery notice, asset record	Satisfaction survey, request status page
What they think	"Where do I order this?"	"Why is so much information needed?"	"Why is my request waiting?"	"Is anyone working on it?"	"Finally, can I get started?"	"That was easier than before."
Pain points	No single place to start; unclear eligibility	Details re-entered that IT already has	Manual approvals, repeated reminders	Manual stock checks, delayed orders	Asset not recorded or assigned on time	Feedback rarely collected
Opportunities with Flow Designer	Single standard laptop catalog item with eligibility check	Pre-fill details from user record; block duplicates	Auto-approve eligible roles; one-click approval; auto-escalate	Auto stock check and purchase order creation	Auto-create asset record; notify at each stage	Auto survey and turnaround dashboard



Key insight: the lowest point is the approval and fulfilment stage, where the employee waits with no visibility. Automating approvals, stock checks, purchase orders and status notifications in Flow Designer removes the manual hand-offs that cause the delay.

Figure 1: Customer journey map and emotional journey

3.2 Solution Requirements

No.	Functional requirement	Sub requirements
FR-1	Laptop request	Standard catalog item; pre-filled form; eligibility and duplicate checks
FR-2	Approval management	Automatic approval for pre-approved roles; manager approval by email or mobile; budget check; escalation after 2 days
FR-3	Inventory and fulfilment	Stock check and reservation; automatic purchase order; vendor assignment
FR-4	Notifications and tracking	Stage notifications; status visible on the portal
FR-5	Asset management	Asset record created and assigned on delivery
FR-6	Reporting	Turnaround time dashboard; automated requests and savings report

No.	Non-functional requirement	Description
NFR-1	Usability	A standard laptop can be requested in a few steps from one catalog item
NFR-2	Security	Role-based access control; only authorized users see request and asset data
NFR-3	Reliability	Failed steps are logged and routed to the coordinator; no request is lost
NFR-4	Performance	Routine flow actions complete within seconds without manual delay
NFR-5	Availability	Follows the availability of the ServiceNow instance
NFR-6	Scalability	New models, roles and vendors are added by configuration

3.3 Data Flow Diagram

The diagram shows how data moves from the employee through approval, stock check, purchase order and delivery, and which data stores each process uses.

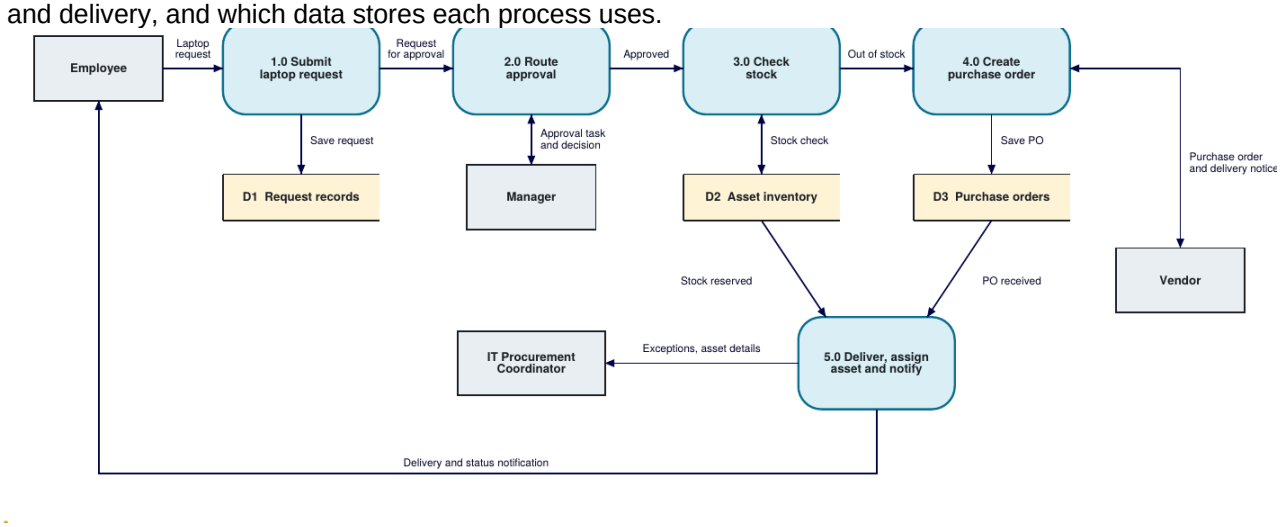


Figure 2: Data flow diagram

3.4 Technology Stack

The solution is built on the ServiceNow platform with Flow Designer as the automation layer.

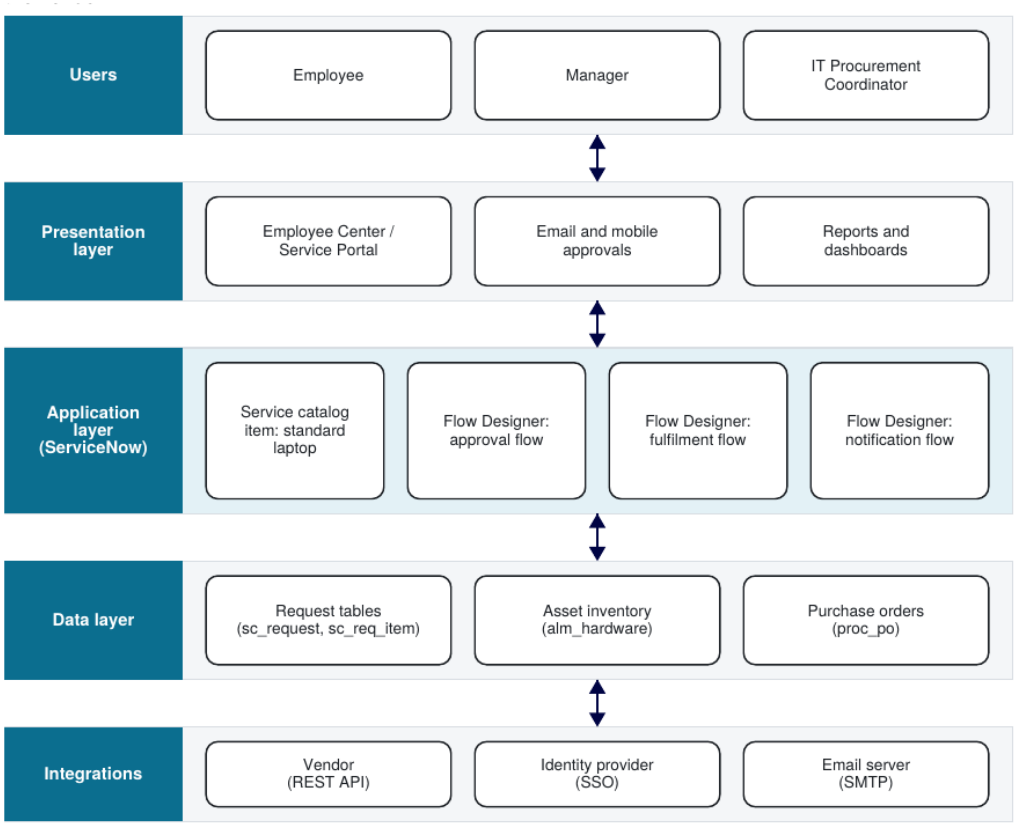


Figure 3: Technical architecture and stack

Component	Technology
User interface	ServiceNow Employee Center / Service Portal
Service catalog	Service Catalog, catalog item and variables
Workflow automation	Flow Designer, subflows and actions
Database	ServiceNow tables (sc_request, sc_req_item, alm_hardware, proc_po)
Notifications	ServiceNow Notifications, email and mobile
External integration	REST Message, IntegrationHub
Reporting	ServiceNow Reports and Performance Analytics

4. PROJECT DESIGN

4.1 Problem Solution Fit

Element	Summary
Customer segments	New and existing employees, approving managers, IT procurement coordinators
Jobs / problems	Get an approved laptop quickly; approve without delay; fulfil accurately. Problems: manual approvals, no status, repeated data entry
Triggers	New hire, refresh cycle, lost or damaged device, team growth
Emotions	Before: anxious, frustrated. After: relieved, confident
Available solutions	Email and spreadsheets; generic catalog with manual approvals; manual stock checks
Constraints	Budget limits, approval and security policies, stock and vendor lead times, limited IT time
Root cause	Manual hand-offs, no automated rules for standard requests, no single view of status
Our solution	Flow Designer automation with a single catalog item, notifications and a dashboard

4.2 Proposed Solution

Parameter	Description
Problem statement	Manual hand-offs delay laptop orders and waste IT time on repetitive work
Idea / solution	Automate the order in ServiceNow with Flow Designer, from request to asset record and notifications
Novelty	Request, approval, inventory, purchasing and asset records connected in one low-code flow, with rules changeable by configuration
Social impact / satisfaction	Faster devices, visible status, less routine work for managers and IT
Business model	Internal efficiency solution on the existing ServiceNow platform; value from shorter turnaround and less manual effort
Scalability	Handles growth in volume and can be reused for other standard hardware

4.3 Solution Architecture

The architecture shows the employee, manager and coordinator working in the ServiceNow platform, where the service catalog starts the Flow Designer flow. The flow uses the platform tables and connects to the identity provider, email server and vendor.

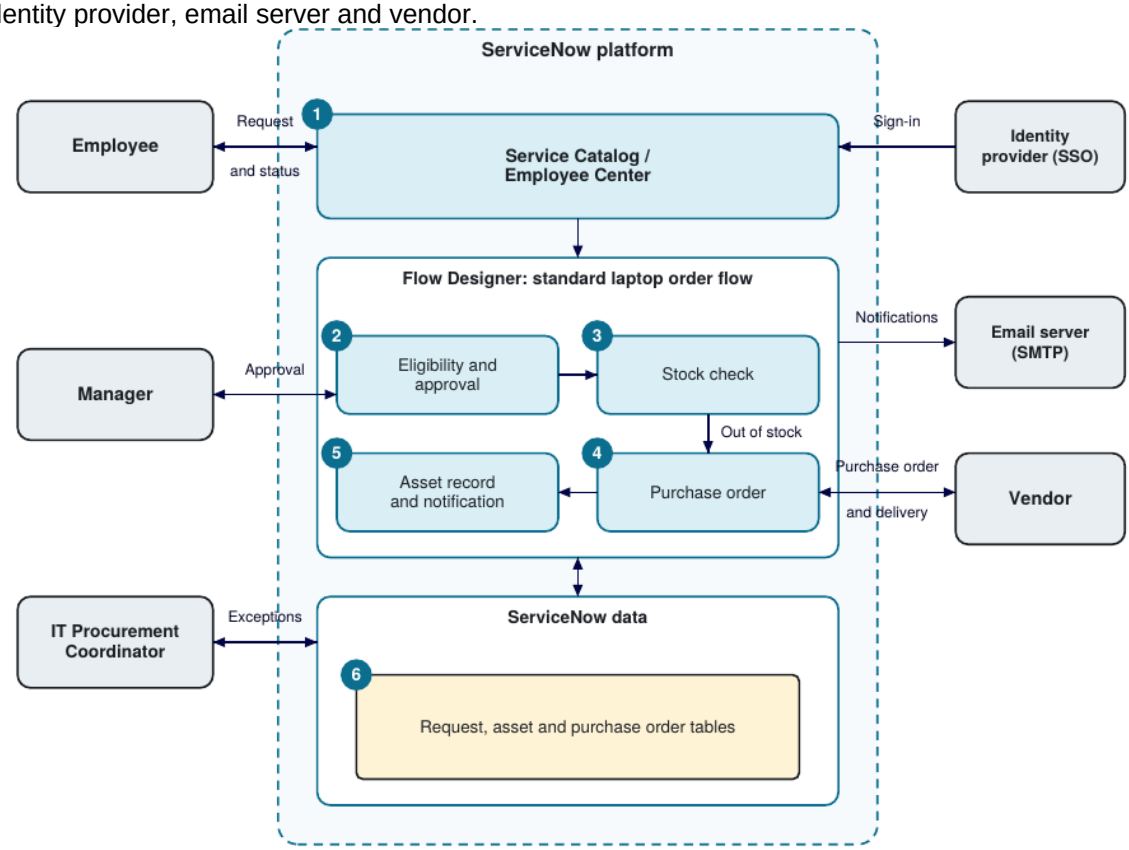


Figure 4: Solution architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The 11 user stories are split into three sprints of 6 days each, estimated in story points. The plan totals 52 story points, an average velocity of about 17.3 points per sprint or 2.9 points per day.

Sprint	Stories	Story points	Dates (planned)
Sprint-1	USN-10 configuration, USN-1 laptop request, USN-4 manager approval	15	03 Feb to 08 Feb 2025
Sprint-2	USN-2 tracking, USN-3 notifications, USN-5 auto-approval, USN-6 stock check, USN-7 purchase order	21	10 Feb to 15 Feb 2025

Sprint	Stories	Story points	Dates (planned)
Sprint-3	USN-8 asset record, USN-9 dashboard, USN-11 testing and release	16	17 Feb to 22 Feb 2025

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional tests check each step of the flow, and performance tests check success rate, execution time and load. User acceptance testing was run with the employee, manager and coordinator roles.

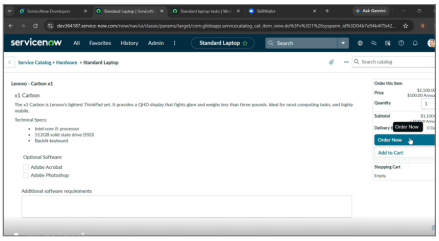
ID	Scenario	Expected result	Result
FT-01	Request form validation	Valid requests accepted; errors for invalid ones	Pass
FT-02	Number and duplicate validation	Out-of-range quantity and duplicates rejected	Pass
FT-03	Approval and fulfilment flow	Correct route for approval and stock outcomes	Pass
FT-04	Vendor and email integration	Vendor responds; email delivered	Pass
PT-01	Flow execution time	Under 5 seconds (average 4.2 seconds)	Pass
PT-02	50 concurrent requests	No noticeable slowdown (98% success)	Pass
PT-03	100 bulk requests	Works without crashing (97 of 100 completed, 3 retried)	Pass

UAT summary: 6 test cases, all passed. 3 bugs were logged: 2 fixed and closed, and 1 (approval button on small mobile screens) still in progress.

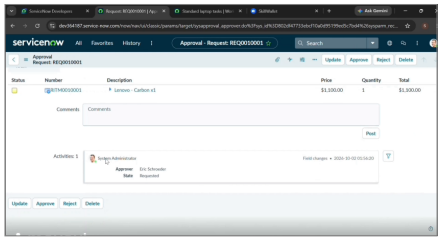
7. RESULTS

7.1 Output Screenshots

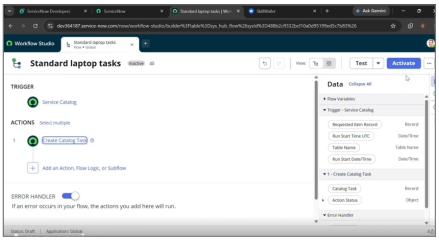
Add screenshots of the working solution below.



Standard laptop request form



Manager approval (ServiceNow approval record)



Flow Designer flow

Add screenshot
Turnaround and savings dashboard

8. ADVANTAGES & DISADVANTAGES

Advantages

- Faster laptop delivery, because routine approvals and stock checks run automatically.
- Less manual work and fewer errors for managers and IT staff.
- Clear request status and notifications for employees.
- Low-code, so rules and vendors can be changed by configuration.

- Built on the existing ServiceNow platform, with no new tools to buy.

Disadvantages

- Depends on the ServiceNow platform and its availability.
- Needs accurate asset inventory and user data to work well.
- Only handles standard laptops, so non-standard requests still need manual handling.
- The vendor integration needs maintenance if the vendor changes its endpoint.

9. CONCLUSION

The project shows that a standard laptop order can be automated end to end in ServiceNow with Flow Designer. The solution addresses the problems found with the people involved: waiting on manual approvals, repeated data entry and no visibility of status. In testing, the flow handled requests reliably and quickly, and user acceptance testing passed all planned cases. The remaining open issue is the mobile approval button.

10. FUTURE SCOPE

- Extend the flow to other standard hardware such as monitors, phones and accessories.
- Compare vendor prices automatically before creating a purchase order.
- Add a mobile-friendly approval screen and chatbot-based requests.
- Forecast stock needs from hiring plans and refresh cycles.

11. APPENDIX

Source Code: The solution is built with low-code Flow Designer flows, subflows and a catalog item rather than written code. Add an update set export or screenshots of the flows here if required.
Dataset Link: Not applicable. The solution uses ServiceNow tables with sample records. [add link if required]
GitHub & Project Demo Link: [add GitHub link here] and [add demo link here]

Note: the dates, test results, counts and figures in this report are sample entries carried over from the earlier project documents. Replace them with your real results before you submit.